Venteicher 2017 IDH-mut astrocytoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9 samples intermix per-neighbourhood (iLISI 1.0→4.99/9) on one continuous manifold (single kNN component, 0 islands); no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

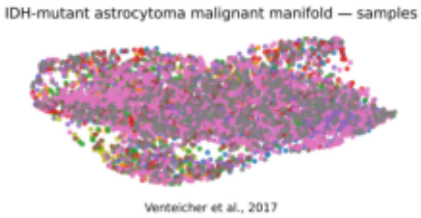
Gast/pSom co-vary (r=+0.60) and both oppose Crest (−0.89 / −0.76): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 — cross-repression → NA

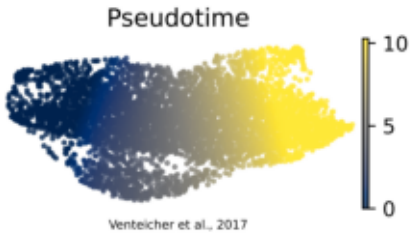
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

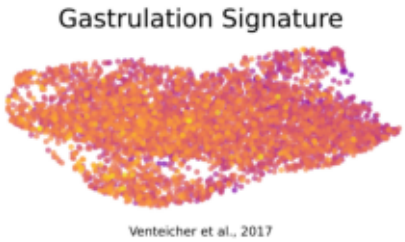
Manifold — samples



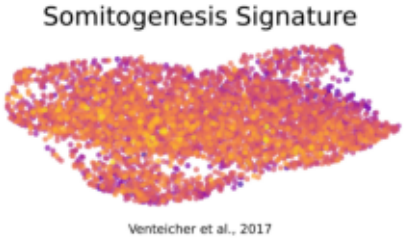
Pseudotime



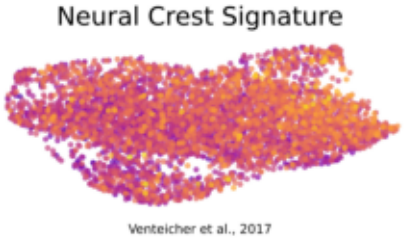
Gastrulation



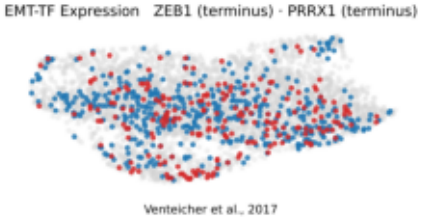
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
IDH-mutant astrocytoma (Venteicher et al., 2017), SmartSeq2 (TPM) — 10 samples; 4,985 malignant cells across 9/10 samples enter the analysis; the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic-program pattern reproduces as the non-branching case: a single continuous gastrulation/somitogenesis→neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Linear pSom/Crest geometry (pSom μ=+0.21 globally strongest). Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. EMT-TFs ZEB1 (91% pan-tumour) → PRRX1 (terminus-enriched).

Parameter choices

Min-cell threshold	≥146 → 9/10 samples, 4,985 cells (tail gap 192→112, r=1.71; geo-mean cut)	**Genes / embedding**	6,317 overdispersed (GAM); n_pcs=20 (scree elbow~6 rejected); Harmony θ=10
Islands removed	0 (single connected kNN component; radius-graph = 1 component)	**Geometry**	Non-branching — scFates curve N=20, λ=0.3, μ=100 (2 tips, 0 forks; fold-free, self-approach 2.96)
Root	curve tip 0 — Gast/pSom-high, Crest-low (developmental origin; Crest is late)	**EMT-TFs**	ZEB1 (91% pan-tumour) → PRRX1 (terminus-enriched); Snail/Twist/FoxC families near-absent